Should-Costing Accuracy Audit
Casting + Machining and Forging + Machining

Technical Audit Report — deliverable A (brief Phase 11-A)

Application	CostVision, repository leamington-marathi, app `calculator/`, commit 7d74ff1b
Date	10 Nov 2026
Costing basis	UK rate book 2.2.0 (2026-09-29); 200,000 parts/yr × 5 yr; EXW; overhead 12 %, margin 8 % (both sides)
Method	Code read + probes through the product's own headless chain (`costMeasuredPart`, `AIR_GAPPED=1`); no repository file was changed
Reference (B)	An independent bottom-up engineering estimate built without reading the application's code. It is **not a market price**; no supplier quote or purchase price was available for any part.

Readiness: Suitable for controlled internal testing

Scope: casting + machining and forging + machining of prismatic / near-net parts, UK book, with the controls listed — not bevel or spiral-bevel gears, which are out of scope (CF-21). Accuracy is **not validated**: no part has been compared with a real price. See section 16 for the justification and the controls.

Headline findings

Finding	Evidence
The arithmetic is deterministic, traceable and re-adds.	Baseline reproduced to the penny on 6 parts; every material bucket re-adds to < £0.0001 (probe P7); overhead and margin applied once.
The tool is above the independent estimate on all 5 parts.	Casting bracket £9.98 v £8.36 (+19.4 %); Part1 £13.82 v £11.87 (+16.4 %); PRCR002 £25.79 v £16.20 (+59.2 %); Steering knuckle RH £12.53 v £9.16 (+36.8 %); Hub flange £13.37 v £10.44 (+28.1 %).
The gap is explained driver by driver; most of it is assumption v assumption.	Σ gap £19.46: secondary-operation constants +£6.66, non-cutting time +£5.23 (A's fixed per-feature allowances alone are £8.94 of its cost), OEE / operator basis +£2.06, cutting time +£1.45, tool wear +£1.31, overhead + margin +£3.36; B-only inspection and scrap −£1.08.
Machined surfaces, machining allowance, setups and the raw-to-finished model are not drawing-driven.	Critical gaps CF-01 to CF-04; single-gap probes move a part by up to −39.5 % (machined-face set, modelled yoke).
No bevel-gear capability.	Historic CH-47 gear: recorded ÷ tool labour hours 14× (forged route) and 61× (conventional); forged billet within +9 %.

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1. Scope, parts and conditions

Three kinds of evidence are used. They must not be mixed up:

Evidence	What it is	What it can show
A — current application	The tool's headless chain on recorded OCP geometry of each STEP (current/*.json).	What the tool does, line by line.
B — independent reference	A bottom-up estimate by a separate engineer with the same rate-book ids (reference/*.json), ISO 8062:1994 allowances, maker cutting data (Sandvik S2–S5, S12), sourced energy figures; many cycle, crew and yield inputs are labelled assumptions.	Which methodology differences drive A − B. Not accuracy: B is not a market price.
Historic record	TRW ER-7389-F (DTIC AD-747857, 1972, Distribution A): CH-47 spiral bevel gear route and standard hours, Tables XXV / XXVI; tool run on STEP models MODELLED FROM THE PUBLISHED DRAWING.	Operation coverage, set-up structure and order of magnitude of hours. Not £.

Part	Status	Route compared (A and B)	A runs used
Casting bracket	genuine CAD (cad-audit/parts)	cast + machine	Casting_Braket__LIKE_B_gravity_LM25_T6_2imp_200k_5y_EXW.json
Part1 (triple clamp)	genuine CAD (user's gravity-die part)	cast + machine	Part1__LIKE_B_gravity_LM25_T6_2imp_200k_5y_EXW.json
PRCR002 (stub axle)	genuine CAD	cast + machine	PRCR002__200k_5y_EXW.json
Steering knuckle RH	genuine CAD	cast + machine (B route); forging recorded	steering_knuckle_RH__LIKE_B_cam_GJS400_200k_5y_EXW.json
Hub flange (MODELLED)	MODELLED in OCP — not a customer part	forge + machine	MODELLED_FORGE_Hub_Flange__200k_5y_EXW.json

> Like-for-like re-runs: where A had chosen a different process or grade from B, A was re-run with B's answers where the tool offers the question (bracket: aluminium LM25, gravity, T6, 2 impressions — the recorded A run was steel sand £9.81; Part1: 2 impressions — the default 1-impression run was £14.60; knuckle: cast + machine EN-GJS-400-15 — the recorded A run was a forging £18.74). PRCR002 and the hub flange were already like for like. Both runs of each are kept in current/.

2. Current application workflow

#	Step	Code (calculator/)	What it does for a cast / forged + machined part
1	Upload	server/routes/cad.ts:773-776	One CAD file (cadFile), one drawing (drawingPdf), one blank DXF.
2	Geometry kernel	server/utils/cad-geometry-engine.py (featureTable :330-449, setup count :1933-2006, turning, draft)	Volume, area, face types, holes with axes, the 8 largest planar areas as "face" rows, pockets, bosses, turning signature, draft, setup directions.
3	Drawing reader (optional)	server/drawing/*, src/engine/engineering-data/to-costing.ts, commodity/*.ts	Reads title block, material, tolerances, GD&T, Ra, threads; only confirmed material, tolerance class, safety, pressure-tight, finish, temper become answers.
4	Route and questions	cost-input-rules/derive/commodity.ts; casting.ts / forging.ts advise()	Commodity from answer or inference; blocking questions (family, pressure-tight, tolerance class, safety).
5	Rules	cost-input-rules/commodities/{casting,forging,machining}.ts; secondary-machining.ts:75-136	Process, yield, cores, line, impressions, die / pattern, fettling, heat treat, NDT; the near-net machining plan (nearNetCut → routingFor → buildOperationPlan).
6	Raw weight	machining-time.ts:553-576 asCastWeight; forging.ts:590-623	Finished + drilled holes + uniform stock on face rows and cored bores (+ draft, turning stock).

#	Step	Code (calculator/)	What it does for a cast / forged + machined part
7	Guards	server/routes/cad.ts runAllGuards; cad-machining-guard.ts	Sanity and near-net cap (caps AI times only).
8	Parameters	cost-input-rules/to-cost-params.ts:447-690	Maps rules to module inputs; casting / forging routes call nearNetSecondaryMachining.
9	Module	modules/casting.ts:160-489; modules/forging.ts:123-342; modules/cast-and-machine.ts	Operations, material, consumables, energy kWh, tooling items.
10	Stack	src/engine/core.ts:150-338	Material = gross × £/kg − credit + consumables + energy; op £ = rate × time ÷ OEE (machine), × crew ÷ efficiency (labour); overhead on factory base; margin on subtotal.
11	Headless chain	server/services/bulk-run.ts:269-425 costMeasuredPart	The chain used for every probe in this audit (identical to the screen per the project's parity tests; not re-verified on screen here).

3. Existing calculation methodology — assessment of 19 areas

Assessment scale from the brief. "Verified" means a probe or a hand re-calculation was run, not that a comment or a test name says so. Full code references, formulas, inputs and evidence: audit/assessment.json; every gap in workbook sheet 11 (Defect & CA register).

Area	Phase	Assessment	Key evidence
As-cast / as-forged drawing interpretation	1, 2	**Missing**	One drawing per costing; no raw-condition stage or allowance parameter exists.
Final machined drawing interpretation	1, 2	**Partially implemented**	Dimensions, GD&T, Ra, threads are read; only material, tolerance class, safety, pressure-tight, finish, temper reach the cost.
Raw and finished CAD geometry extraction	1, 2	**Partially implemented**	Finished STEP measured by OCCT (verified by re-reading the files); no raw-CAD input.
Raw and finished component weight	1, 3, 8	**Partially implemented**	Finished = volume × density (verified). Raw = finished + uniform stock + drilled holes (+ draft on forgings); hand re-calc 3.039 kg matches.
Material and manufacturing process identification	1, 7	**Partially implemented**	Answers or a representative grade; advisor picks the process. Route was an engineer answer in every case tested.
Machining allowance calculation	1, 3	**Incorrect / unreliable**	One per-side constant per process (cast) or per size band (IS 3469, forged); time does not move with stock (P4).
Machining feature recognition	1, 2, 4	**Incorrect / unreliable**	Holes good (stepped bores, lands). Machined faces = 8 largest planar areas incl. drafted walls (yoke 22 v 4; −39.5 %).
Machining operation sequencing	1, 4	**Missing**	No OP10/OP20 or datums; one operation per kernel direction, time shared by ALL faces in that direction.
CNC machine and axis selection	1, 5	**Incorrect / unreliable**	3 routings priced on £; setups the geometry cannot achieve (P3); 5-axis assumed 1 setup.
Number of setups and fixture requirements	1, 5	**Incorrect / unreliable**	Kernel count ≤ 3 by construction; 4 of 6 parts need ≥ 2 clampings; 2 clampings add +2.5 to +5.0 %.
Setup time and machining cycle time	1, 6	**Partially implemented**	Setup separate and amortised (verified); cutting data sourced; fixed per-feature allowances 28–76 % of minutes (P10).
Tooling, labour, machine rates and overheads	1, 8	**Partially implemented**	Overhead 12 % and margin 8 % applied once (re-added); bought-in services take them again.
Material consumption, scrap, yield and recovery	1, 7, 8	**Partially implemented**	Casting balance closes; forging 25.7 % of billet un-itemised; swarf never credited; machining scrap dropped.
Inspection and secondary operations	1, 7, 8	**Partially implemented**	0.5-min gauge check; NDT only when safety-critical; riser cut-off and weld repair not costed.
Final should-cost calculation	1, 8	**Partially implemented**	Deterministic; reproduces the recorded baseline to the penny; inherits the input gaps.

Area	Phase	Assessment	Key evidence
Casting primary-process model (separate from forging)	7A	**Partially implemented**	Separate module; DISA / BREF sourced items; yield and cores are bands; gating not designed.
Forging primary-process model (separate from casting)	7B	**Partially implemented**	Separate module; saw, heater, press, trim, dies; flash fixed 10 %, yield band unsourced.
Double counting between primary process and machining	7, 8	**Implemented and verified**	None found; machining scrap is under-counted instead.
Transparent cost bridge raw material → finished	8	**Partially implemented**	Trace re-adds to £0.0001, but the material bucket mixes metal with services.

4. Casting + machining assessment

Element (brief 7A)	Existing behaviour and basis	Assessment
Material input and charge	Metal charge £/kg from the rate book; bought = as-cast + melt loss on remelted returns (modules/casting.ts:160-185).	Implemented and verified (closes, P7)
Casting process selection	Advisor on alloy, mass, volume, tolerance class, safety; process can be answered (casting.processChoice).	Partially implemented
Casting yield	Band midpoint by process / alloy (sand steel 0.525, ductile 0.625, gravity Al 0.6); not computed.	Partially implemented (CF-10)
Gates, runners, risers	Not designed; inside the yield band; remelted, 2–3 % lost.	Missing (CF-10)
Moulds, cores and tooling	DISA line by moulds/yr; impressions by the DISA 60 % plate rule (sourced); pattern / die toolroom build-up (judgement); cores = £0.50–8 band by undercut-face count.	Partially implemented (CF-11)
Melting, pouring, cooling	Melt kWh on every kg poured; ladle preheat and dust from EU BREF; plant electricity a grade-D estimate; melt crew 1.5 op-h/t.	Partially implemented
Fettling, trimming, cleaning	Robot cell 40 s (one time study) for ferrous ≤ 30 kg; 2 min flat bench for gravity Al; blast by throughput heuristic or £0.10 floor; riser cut-off and weld repair on steel NOT costed.	Partially implemented (CF-13)
Heat treatment	Steel normalise £0.35/kg (unsourced); Al T6 / T5 station ESTIMATE (£0.41/kg on the bracket).	Partially implemented (CF-13)
Inspection and rejection	3 % casting reject uplifts line, pattern and consumables; MPI / RT only if safety-critical.	Partially implemented (CF-12)
Subsequent machining	Near-net plan (secondary-machining.ts) shared by casting, cast + machine and forging routes; machining scrap dropped from the join.	Partially implemented (CF-01, CF-05, CF-06, CF-09, CF-19)

Run evidence: bracket (gravity LM25 T6, 2 impressions) £9.98; Part1 £13.82; PRCR002 (sand GJS-500-7) £25.79; knuckle (sand GJS-400-15) £12.53 — 200k/yr × 5 yr, EXW, UK. Operation lists in workbook sheet 5.

5. Forging + machining assessment

Element (brief 7B)	Existing behaviour and basis	Assessment
Billet or bar input	Billet = (as-forged + flash) ÷ yieldFraction = as-forged ÷ 0.675 (unsourced band); bar = billet + saw kerf.	Incorrect / unreliable (CF-04)
Billet heating	Induction 0.4 kWh/kg on every billet (maker claim, SMS Elotherm); heater time by kg/h.	Implemented
Process and equipment	Advisor process; press by force on a ladder; takt 3 s + hits × 4 s (unsourced); crew 2 + trim 1.	Partially implemented
Dies and tooling	Toolroom build-up ÷ die life (38,000 forgings, unsourced), fractional sets; no die-change set-up.	Partially implemented (CF-22)

Element (brief 7B)	Existing behaviour and basis	Assessment
Flash and trimming	Flash fixed at 10 % of the forging; trim press sized from a flash line computed for force only.	Incorrect / unreliable (CF-04)
Scale and material losses	Not modelled: 25.7 % of every billet is "scale, crop, wad (not itemised)" and credited at the steel scrap price.	Incorrect / unreliable (CF-04, CF-14)
Heat treatment	By grade (38MnVS6 controlled-cooled, none); quench-temper £/kg for others.	Partially implemented
Shot blasting and finishing	Descale £0.12/kg × forging (unsourced).	Partially implemented (CF-13)
Inspection and rejection	2 % forging reject; MPI 75 s station if safety-critical.	Partially implemented (CF-12)
Subsequent machining	Same near-net plan as castings; pierced bores > Ø25 finish-bored; IS 3469 finish stock.	Partially implemented (CF-01, CF-02, CF-07)

Run evidence: hub flange (MODELLED) £13.37 at 200k/yr × 5 yr; knuckle as a forging £18.74 (route note). Historic CH-47 gear: the forging route picked a ring mill (CF-21).

6. Raw-to-finished drawing interpretation (brief Phase 2)

Brief Phase 2 capability	Can the application do it?	Evidence
1. Identify and associate the drawing pair	No	One drawing per costing (cad.ts:773-776; drawingHash singular).
2. Verify part number, revision, units, identity	No (units partly)	title.partNumber / title.revision read (title-block.ts:18-26), never compared with the CAD; title.units used only for tolerance interpretation.
3. Extract raw-condition dimensions and geometry	No	No stage parameter in the dictionary; no raw-CAD input.
4. Extract finished dimensions, tolerances, GD&T	Read but not used	dimension, gdt.frame, gdt.datum, finish.roughness are extracted; no costing binding (engineering-data/commodity/*.ts).
5. Identify machined and unmachined surfaces	Heuristic only	8 largest planar areas + every hole (CF-01).
6. Features created, enlarged or finished by machining	Partly	Holes ≤ Ø20 (cast) / ≤ Ø25 (forged) treated as drilled from solid; larger as cored / pierced and finish-bored.
7. Reconcile dimensions, detect inconsistencies	No	No raw v finished comparison exists; no "insufficient stock" check.
8. Traceability to drawing and CAD source	Partly	Review decisions carry drawing hash and location; costing traces carry rule bases.

What the genuine raw / machined pairs show

Four genuine pairs were found in public sources (dataset/register.json). None comes with CAD, so none can be run through the tool.

Pair	Route	"Raw" drawing	Finished drawing	CAD	Real cost data
Crankcase, Automotive Stirling Engine Mod I (U-4 engine drive system)	casting + machining	CRANKCASE (sheets 1-7 of 7) 1-61116 (internal no. 3274-116) — CAST-PART DETAIL drawing (cast geometry + machining done on the single part); NOT a pure as-cast drawing	CRANKCASE/BEDPLATE MACHINING DRAWING (sheets 1 and 2 of 2)	none	no
Bedplate (lower crankcase), Automotive Stirling Engine Mod I	casting + machining	BEDPLATE (sheets 1-4 of 4) 1-61117 (internal no. 3274-117) — CAST-PART DETAIL drawing (cast geometry + machining done on the single part); NOT a pure as-cast drawing	CRANKCASE/BEDPLATE MACHINING DRAWING (sheets 1 and 2 of 2)	none	no

Pair	Route	"Raw" drawing	Finished drawing	CAD	Real cost data
Spiral bevel GEAR, 43 teeth, CH-47C helicopter nose gearbox (precision-forged teeth)	forging + machining	Figure 5. Gear forging design SK22270 / FR529997 (forging design) — as-forged (forging design: forging stock 0.006-0.010 at MMC on teeth, 'STOCK ALLOWANCE @ MMC SEE DETAIL B', 9-3/4 approx dia)	Figure 45. Gear machining sequence sketches and layout (route card + operation sheets) and Table I Basic Gear and Pinion	none	yes (1972 standard hours, Tables XXV / XXVI)
Spiral bevel PINION, 35 teeth, integral shaft, CH-47C helicopter nose gearbox input	forging + machining	Figure 4. Pinion forging design SK22269 / FR729996 (forging design) — as-forged (0.006/0.010 stock allowance on teeth to MMC of finish part; shank 4.015/3.985 dia, 8 in long; head 8 in dia approx)	Figure 44. Pinion machining sequence sketches and layout (route card for job-cost orders FR-729982-4 'Forged pinion gear	none	no

> The Stirling crankcase and bedplate "raw" drawings are cast-part detail drawings (cast geometry plus machining on the single part), not pure as-cast drawings; the finished drawing is an assembly-level line-boring drawing. Only the CH-47 forging designs are true as-forged drawings. Conclusion: even with genuine pairs, a raw-to-finished model needs both drawings read AND a CAD — the tool has neither the second drawing input nor a raw-geometry input. Where the CAD is finished-only (all 6 CAD parts here), the raw shape is derived by adding a uniform stock; it is not measured.

7. Machining allowance accuracy (brief Phase 3)

Part	A stock a side	A raw	B stock a side and basis	B raw	Finished
Casting bracket	1.5 mm uniform (gravity)	0.965 kg	1.55 mm — largest finished dimension 132.6 mm → band 100–160; gravity die light metal RMAG D–F (Table B.1) → E = 1.1 mm; CT: Table A.1 gives no data for permanent mould → CT8 = 1.8 mm (assumption)	0.926 kg	0.858 kg
Part1 (triple clamp)	1.5 mm uniform (gravity)	1.348 kg	2.35 mm — largest 270 mm → band 250–400; RMAG E = 1.8 mm; CT8 = 2.2 mm (assumption) → 2.35 mm/side	1.399 kg	1.151 kg
PRCR002 (stub axle)	3 mm uniform (sand)	9.084 kg	3.6 mm — largest 274.5 mm → band 250–400; machine-moulded SG iron RMAG E–G → F = 2.5 mm; CT8–12 (Table A.1) → CT10 = 4.4 mm → 3.6 mm/side	9.278 kg	7.364 kg
Steering knuckle RH	3 mm uniform (sand)	3.238 kg	3 mm — largest 236.4 mm → band 160–250; RMAG F = 2.0 mm; CT10 = 4.0 mm → 3.0 mm/side	3.096 kg	2.528 kg
Hub flange (MODELLED)	1.5 mm uniform (closed-die)	3.185 kg	2.2 mm — 1.5 mm per machined surface (S7, 0.06 in) + 0.7 mm = minus half of an ESTIMATED grade-F tolerance (+1.4/−0.7 mm for a ~3.4 kg, 145 mm, M1, S2 forging — EN 10243-1 Table 1 is an image and was NOT read; the S6 worked example gives +1.1/−0.5 at 0.2 kg, 78 mm). Tolerance kept separate: the allowance is nominal stock, the forging may vary +1.4/−0.7 about it; scale pits ≤ ½ allowance (S6 §5.2.4).	3.425 kg	2.197 kg

Question	Finding
Per surface / feature or uniform?	Uniform: one figure for every face row and cored bore of the part (machining-time.ts:379-390, 531-538).
Casting / forging tolerance v machining allowance?	Not distinguished. The casting basis text states the figure should be RMA + ½ DCTG but neither is read. Forgings use IS 3469 minimum finish stock; EN 10243-1 tolerances are not added.
Roughing v finishing stock?	Not distinguished. One finishing face-mill pass and one boring pass whatever the stock (CF-06). Only the near-net turning time grows with stock.
Size dependence?	Castings none; forgings by greatest dimension (IS 3469 Table 1.5). B uses ISO 8062 RMA by largest dimension: 1.55 mm (133 mm part) to 3.6 mm (275 mm part).
Insufficient stock / conflicts flagged?	No.

Question	Finding
Cost sensitivity	Casting stock 3.0 → 1.5 mm a side, Casting bracket (steel sand, 50k): −£0.31 (−2.9 %); Casting stock 3.0 → 6.0 mm a side, Casting bracket (steel sand, 50k): +£0.63 (+5.8 %); Casting stock 3.0 → 1.5 mm a side, PRCR002 (50k): −£0.88 (−3.2 %); Casting stock 3.0 → 6.0 mm a side, PRCR002 (50k): +£1.77 (+6.5 %); Forging stock 2.5 → 1.5 mm a side, Knuckle forging (50k): −£0.38 (−2.0 %); Forging stock 2.5 → 4.0 mm a side, Knuckle forging (50k): +£0.59 (+3.0 %). Machining-centre minutes unchanged in every case.

8. Operation sequence assessment (brief Phase 4)

Part	A operation lines (no OP numbers, no datums)	B operations	Probe P3 (machined features)
Casting bracket	7 lines: Machining Setup; Finish machining — +Z side; Finish machining — +X side; Finish machining — +Y side; Drilling — 6 holes; Pallet change — 1 fixturing(s); Deburr and gauge check	OP10 mach-hmc-400 (7 features); OP20 mach-hmc-400 (5 features)	7 directions; ≥ 2 clamping(s) on one rotary axis
Part1 (triple clamp)	8 lines: Machining Setup; Finish machining — +X side; Finish machining — +Z side; Finish machining — +Y side; Drilling — 2 holes; Finish boring — 5 cored bore(s), 3 stepped; Pallet change — 1 fixturing(s); Deburr and gauge check	OP10 mach-hmc-400 (12 features); OP20 mach-hmc-400 (3 features)	11 directions; ≥ 2 clamping(s) on one rotary axis
PRCR002 (stub axle)	10 lines: Machining Setup; Finish machining — +Y side; Finish machining — +X side; Finish machining — +Z side; Drilling — 12 holes; Finish boring — 3 cored bore(s), 3 stepped; Turning — spindle on the CNC lathe; Load / clamp / unload — lathe fixture, 1 fixturing; Pallet change — 1 fixturing(s); Deburr and gauge check	OP10 mach-hmc-400 (17 features); OP20 mach-lathe-cnc (2 features); OP30 mach-hmc-400 (6 features)	8 directions; ≥ 2 clamping(s) on one rotary axis
Steering knuckle RH	10 lines: Machining Setup; Finish machining — +X side; Finish machining — +Z side; Finish machining — +Y side; Drilling — 7 holes; Finish boring — 1 cored bore(s), 1 stepped; Turning — spindle on the CNC lathe; Load / clamp / unload — lathe fixture, 1 fixturing; Pallet change — 1 fixturing(s); Deburr and gauge check	OP10 mach-hmc-400 (21 features); OP20 mach-hmc-400 (11 features)	9 directions; ≥ 2 clamping(s) on one rotary axis
Hub flange (MODELLED)	7 lines: Machining Setup; Drilling — 6 holes; Finish boring — 1 cored bore(s); Turning — outside and faces on the CNC lathe, op 10 / op 20; Load / clamp / unload — lathe fixture, 1 fixturing; Load / clamp / unload — 1 fixturing(s); Deburr and gauge check	OP10 mach-lathe-cnc (2 features); OP20 mach-lathe-cnc (2 features); OP30 mach-vmc3 (2 features)	3 directions; ≥ 1 clamping(s) on one rotary axis

- A has no sequence: a set of lines — one "Finish machining — <direction> side" per kernel direction, whose time is the total face-milling time shared by the count of **all** B-rep faces in that direction (bracket +Z / +X / +Y = 1.48 / 0.95 / 0.71 min from 108 / 70 / 52 of 230 faces at 50k), plus one drilling, one boring, handling and deburr line.
- No tool list per operation, no datum, no tolerance relationship. The hub flange's Ø60 bore is finish-bored on the 5-axis mill apart from the lathe operation that turns the OD (concentricity datum broken); the lathe "op 10 / op 20" is charged one load.
- B builds OP10 / OP20 (/ OP30) from datums (cast datum targets → machined face A + bores), with workholding and orientation per operation.

9. CNC machine and axis selection (brief Phase 5)

Part	A chosen	A alternatives priced (machine + operator £/part)	Directions / one clamping feasible?	B machines
Casting bracket	volume-cell	volume-cell £1.75 / consolidated-5axis £2.29 / split-3axis £2.79	7 / no	OP10: mach-hmc-400; OP20: mach-hmc-400
Part1 (triple clamp)	volume-cell	volume-cell £2.89 / consolidated-5axis £3.51 / split-3axis £3.72	11 / no	OP10: mach-hmc-400; OP20: mach-hmc-400
PRCR002 (stub axle)	volume-cell	volume-cell £4.60 / consolidated-5axis £5.78 / split-3axis £6.46	8 / no	OP10: mach-hmc-400; OP20: mach-lathe-cnc; OP30: mach-hmc-400

Part	A chosen	A alternatives priced (machine + operator £/part)	Directions / one clamping feasible?	B machines
Steering knuckle RH	volume-cell	volume-cell £2.85 / consolidated-5axis £3.46 / split-3axis £3.66	9 / no	OP10: mach-hmc-400; OP20: mach-hmc-400
Hub flange (MODELLED)	consolidated 5-axis + lathe	not printed for the forging route; 5-axis UMC-500 + CNC lathe chosen	3 / yes	OP10: mach-lathe-cnc; OP20: mach-lathe-cnc; OP30: mach-vmc3

- **Setup count.** The kernel clusters every B-rep face normal, merges +N with −N and keeps only +X / +Y / +Z labels (cad-geometry-engine.py:1933-2006), so the count is ≤ 3 by construction. The HMC cell then puts ≤ 4 directions in 1 clamping (routing-optimiser.ts:303) and the 5-axis option is 1 setup on a near-net part (:295) — neither checks that one rotary axis can reach the directions.
- **Alternatives are compared** on machine + operator £, which the brief asks for; but on setup counts the geometry cannot achieve (P3: 4 of 6 parts need ≥ 2 clampings). Envelope is checked; part weight, fixture complexity and tolerance capability are not.
- **Effect of 2 clampings** (P4-D, 50k): Casting bracket +£0.39 (+3.6 %); PRCR002 +£0.68 (+2.5 %); Knuckle forging +£0.97 (+5.0 %); Part1 (machining) +£1.27 (+3.3 %).

10. Setup and cycle-time assessment (brief Phase 6)

Part	A machining	of which fixed allowances	A handling	B cycle	B cut	B non-cut	A setup / part
Casting bracket	3.67 min	2.62 min (71 %)	0.12 min	1.74 min	0.24 min	1.50 min	0.54 s
Part1 (triple clamp)	6.02 min	4.55 min (76 %)	0.12 min	3.35 min	1.05 min	2.30 min	0.54 s
PRCR002 (stub axle)	12.46 min	4.51 min (36 %)	1.32 min	8.52 min	4.82 min	3.70 min	1.08 s
Steering knuckle RH	7.37 min	2.95 min (40 %)	0.72 min	6.70 min	3.78 min	2.92 min	1.08 s
Hub flange (MODELLED)	4.11 min	1.15 min (28 %)	1.20 min	4.08 min	2.36 min	1.72 min	1.08 s

Brief Phase 6 item	In the tool
Initial setup and per-operation setup	45 min per fixturing per batch (unsourced), batch = annual ÷ 20 (50–5,000); one amortised line. No die-change or forge set-up (CF-22).
Cycle per operation	Cutting from maker data (cutting-catalogue.ts, sourced) + fixed approach allowances per feature (0.06–0.5 min, unsourced) + 2.2 s tool changes (sourced).
Rapids, probing, first-off	No rapid or probing model; first-off inside the 45 min setup.
Tool wear	£/min on "cutting" minutes that include the allowances (CF-20).
Labour	Cutting crew 0.5, skilled £27.95 ÷ 0.92; handling and bench crew 1.
Machine hourly cost	Rate book, machine only; ÷ OEE 0.8.
Historic check (CH-47 gear)	Recorded ÷ tool labour hours: forged route 13.8× (Table XXIX basis); conventional route 61× labour. Era, method and scope sit inside these ratios.

11. Material balance and yield (brief Phases 7–8)

Part	Finished	Raw part A / B	Metal bought per good part A / B	Removed by machining A / B	A credit	B credit
Casting bracket	0.858 kg	0.965 kg / 0.926 kg	0.993 kg / 0.985 kg	0.107 kg / 0.068 kg	£0.00	£0.04
Part1 (triple clamp)	1.151 kg	1.348 kg / 1.399 kg	1.387 kg / 1.489 kg	0.197 kg / 0.248 kg	£0.00	£0.14
PRCR002 (stub axle)	7.364 kg	9.084 kg / 9.278 kg	9.261 kg / 9.707 kg	1.720 kg / 1.915 kg	£0.00	£0.29
Steering knuckle RH	2.528 kg	3.238 kg / 3.096 kg	3.301 kg / 3.240 kg	0.710 kg / 0.568 kg	£0.00	£0.09

Part	Finished	Raw part A / B	Metal bought per good part A / B	Removed by machining A / B	A credit	B credit
Hub flange (MODELLED)	2.198 kg	3.185 kg / 3.425 kg	4.903 kg / 4.205 kg	0.987 kg / 1.227 kg	£0.33	£0.39

- **Castings (A).** Bought = as-cast + melt loss on remelted returns (2–3 %); gating and rejects remelted; it closes (P7). Swarf is never credited (lossIsNotScrap).
- **Forging (A).** Billet = as-forged ÷ 0.675; 25.7 % of every billet is un-itemised (constant by construction); all losses credited at £0.20/kg including scale; swarf not credited. Hub flange: A billet 4.716 kg v B 3.947 kg (B itemises flash 0.348, wad 0.145, scale 0.030 kg).
- **B** closes every balance to 0.0 g, credits swarf and recoverable streams, and treats dross / scale as non-recoverable.

12. Cost breakdown and calculation gaps (brief Phase 8)

Brief Phase 8 line	In the tool	Gap
1 Raw material input	Gross kg × £/kg in the "rawMaterial" bucket, together with energy, heat treat, NDT, cores, blast, tool wear.	CF-17 (Low)
2 Casting / forging conversion	Per operation (line, melt crew, fettling; saw, heater, press, trim).	—
3 Secondary processes	Consumable £ items (heat treat, blast, descale, NDT) inside material.	CF-13, CF-17
4 Machining cost by operation	Per line; per-side lines apportioned by face count.	CF-08, CF-18
5 Setup and fixture costs	Setup line amortised; fixtures + programming as tooling items.	CF-05
6 Tooling amortisation	Items ÷ annual × years; fractional die sets.	—
7 / 8 Labour and machine	Separate per operation in the trace.	—
9 Inspection and quality	NDT station if safety-critical; 0.5-min gauge check.	CF-12
10 Scrap, yield, recovery	Reject uplifts; casting remelt; forging credit; no swarf credit; no machining scrap.	CF-04, CF-09, CF-14
11 Overheads	12 % of factory base, 8 % margin on subtotal, applied once (verified). Bought-in services carry both.	CF-13

Composition of the tool's "raw material" bucket (P7): metal net of credits is Casting bracket 83 %, Part1 83 %, PRCR002 29 %, Steering knuckle RH 24 %, Hub flange 83 % of it.

13. Component validation results (brief Phase 9)

> B is an independent engineering estimate, not a market price. A − B therefore shows which *methodology* differences drive the result; it is not an accuracy figure for either side. "AvA" marks a driver where both sides' input is an unsourced assumption.

Part	A (tool)	B (reference)	B range	A − B	A − B % of B
Casting bracket	£9.98	£8.36	£7.84 – £9.16	+£1.62	+19.4 %
Part1 (triple clamp)	£13.82	£11.87	£11.16 – £12.94	+£1.95	+16.4 %
PRCR002 (stub axle)	£25.79	£16.20	£15.03 – £17.87	+£9.59	+59.2 %
Steering knuckle RH	£12.53	£9.16	£8.36 – £10.26	+£3.37	+36.8 %
Hub flange (MODELLED)	£13.37	£10.44	£9.95 – £11.20	+£2.93	+28.1 %
All five	**£75.48**	**£56.02**		**+£19.46**	

A is above the top of B's range on every part. Driver tables follow; every table sums exactly to A − B (the residual is B's 3-dp rounding).

Systematic causes across the five parts (pre-overhead except the last row)

Cause	Casting bracket	Part1	PRCR002	Steering knuckle RH	Hub flange	All
Non-cutting time: A's fixed per-feature allowances + handling v B's per-operation non-cut (CF-19)	+£0.79	+£1.59	+£1.55	+£0.68	+£0.62	+£5.23
Secondary-operation constants: cores, fettling, blast, deburr, heat treatment (CF-11, CF-13, CF-16)	+£0.64	+£0.43	+£3.59	+£1.28	+£0.71	+£6.66
Machining OEE 0.8 and skilled operator ÷ 0.92 v B's semi-skilled at 100 % (CF-16)	+£0.13	+£0.16	+£0.90	+£0.50	+£0.37	+£2.06
Tool wear charged on minutes that include the allowances (CF-20)	+£0.13	+£0.19	+£0.63	+£0.29	+£0.07	+£1.31
Material balance, scrap credits, melt / heating energy (CF-02, CF-04, CF-10, CF-14)	+£0.12	−£0.11	−£0.19	+£0.06	+£0.97	+£0.85
Cutting time: feature set, passes, cutting data, lathe op (CF-01, CF-06, CF-07)	+£0.37	+£0.14	+£0.99	+£0.01	−£0.04	+£1.45
Inspection and scrap provisions only B carries (CF-12, CF-09)	−£0.40	−£0.43	+£0.06	+£0.12	−£0.45	−£1.08
Casting / forge cell, primary tooling, fixtures, packaging, rounding	−£0.45	−£0.37	+£0.41	−£0.15	+£0.18	−£0.38
Overhead 12 % and margin 8 % on all of the above (same rates both sides)	+£0.28	+£0.34	+£1.65	+£0.58	+£0.50	+£3.36
Total A − B	**+£1.62**	**+£1.95**	**+£9.59**	**+£3.37**	**+£2.93**	**+£19.46**

Casting bracket — A £9.98 v B £8.36: +£1.62 (+19.4 %)

- A as recorded (baseline ANSWERS: steel, sand GS-C25 on a DISA line): £9.8108 — not comparable with B (aluminium gravity).
- A re-run with B's answers (material.family aluminium, material.grade mat-lm25, casting.processChoice gravity, casting.heatTreat T6, casting.impressions 2): £9.9790 — used below.
- Same with 1 impression (the tool's default): £10.8382.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Metal bought (net of melt loss / incl. bar kerf)	£2.96	£2.91	+£0.05	A as-cast 0.965 kg (1.5 mm gravity stock on 8 face rows = 18,166 mm² + 12.7 cm³ drilled holes) + 3 % loss on returns → 0.993 kg; B raw 0.926 kg (ISO 8062 RMA E 1.1 + CT8/4 = 1.55 mm on 4 faces = 7,444 mm²) + 2.5 % dross of poured → 0.975 kg. Same £2.98/kg.	CF-01, CF-02	B — B uses a size-dependent ISO 8062 allowance on the faces it judges machined; A puts stock on the 8 largest planar areas (CF-01). Neither is checked against a drawing.
Scrap credits (swarf, flash, wad, crop)	£0.00	−£0.04	+£0.04	A credits nothing on a casting (lossIsNotScrap); B credits 0.068 kg swarf at £0.54/kg.	CF-14	B — Aluminium swarf is sold.
Melt / billet-heating energy + foundry indirect (melt-shop fixed, sand, bentonite, sand plant) (AvA)	£0.20	£0.17	+£0.03	A: melt + plant electricity kWh on 1.66 kg poured (yield 0.6) at £0.182; B: 600 kWh/t gas melter (StrikoWestofen, sourced) at £0.046 + holding + £20/t melt-shop fixed + treatment on 1.97 kg poured (yield 0.5).	CF-10	neither — Electric v gas melting is a plant choice; both yields are assumptions.
Casting / forge cell (machine + crew, incl. melt crew) (AvA)	£0.76	£1.12	−£0.37	A die cycle 3.22 min / 2 impressions (mass / section correlation with cited anchors), crew 0.25, OEE 0.8; B 240 s / 2, crew 0.5 + melt crew 1/6, no OEE.	CF-16	neither — Die cycle and crew are assumptions on both sides; B's 240 s is ~25 % longer than A's.
Primary tooling (die / pattern, maintenance, die change) (AvA)	£0.08	£0.16	−£0.09	A gravity die £12,988 (179 toolroom h + 70 kg H13) × 6 sets; B £24,552 (290 h + 260 kg H13 + 2 core pulls) × 6 + £0.03/shot maintenance.	CF-16	neither — Both toolroom build-ups are judgement; B adds die maintenance A omits.
Fettling (AvA)	£0.92	£0.41	+£0.51	A 2 min bench fettling at the foundry rate ÷ 0.92 (flat for gravity aluminium, "engineering-typical, not sourced"); B 45 s + band saw.	CF-16	B — A flat 2 min on a 0.9 kg casting looks high; both unsourced.
Heat treatment (AvA)	£0.39	£0.58	−£0.19	A T6 station ESTIMATE £0.41/kg (0.948 kWh/kg + operator h/kg + capital £0.0078/kg); B furnace hours (8 h solution + 7 h age) × £15.97/h ÷ 400 kg charge = £0.63/kg.	CF-13	neither — No toll price on either side; A's capital charge (£0.0078/kg) looks low.
Shot blast / descale (AvA)	£0.10	£0.04	+£0.06	A £0.10 floor (CostVision heuristic); B 10 min a 150 kg load.	CF-13	neither — Both unsourced.
Inspection and NDT	£0.00	£0.33	−£0.33	A none beyond the 0.5-min gauge check inside "deburr"; B visual 12 s + X-ray 1 in 50 + in-process gauging 15 s + CMM audit 1 in 250.	CF-12	B (partly) — In-process gauging and a CMM audit are normal; an X-ray audit on a non-safety bracket is B's own assumption.
Machining — non-cutting time: A fixed per-feature allowances + handling v B per-operation non-cut (AvA)	£1.48	£0.69	+£0.79	A: 2.62 min of fixed allowances (0.2 min a face, 0.15 a hole, 0.25 a bore pass, 0.1 blind, 0.1 tap; "engineering-typical, not sourced") + 0.12 min handling / pallet change = 2.74 min; B: 1.50 min non-cut (tool changes, approaches, probing, indexing, pallet change). A at B's OEE / operator basis. A 3.67 min on the HMC (8 faces × (0.2 min + one finish pass) + 6 hole works at 0.15–0.25 min approach + 0.1 blind + tap, pallet change 7.1 s) v B 1.74 min (0.24 cut + 1.50 non-cut over OP10/OP20).	CF-19 (new)	B (more plausible) — Both are assumptions; B builds non-cut per operation from tool changes, approaches, probing and indexing; A adds 9–15 s per feature. Neither is a timed cycle.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Machining — cutting time (feature set, passes, cutting data)	£0.48	£0.11	+£0.37	A 0.93 min cutting (incl. 2.2 s tool changes and any lathe op) v B 0.24 min, both at B's basis, split by minutes. A faces its 8 largest planar areas (18,166 mm²) with a finish face mill at the catalogue rate; B faces 4 (7,444 mm²) with a PCD mill at vc 1,000 m/min and drills at vc 200 m/min.	CF-01, CF-06	neither — The face set is not drawing-backed on either side (CF-01); cutting data are sourced on both.
Machining — OEE 0.8 on machine time (A) v none (B) (AvA)	£0.07	£0.00	+£0.07	A divides machine time by OEE 0.8 (core.ts:247); B applies no availability / performance loss.	CF-16	A — A dedicated cell still loses availability; B is optimistic. A's 0.8 is unsourced.
Machining — operator grade and efficiency (AvA)	£0.06	£0.00	+£0.06	A: skilled machinist £27.95 ÷ 0.92 efficiency, 0.5 crew; B: semi-skilled £24.65 at 100 %, 0.5 crew.	CF-16	neither — A production-cell loader is semi-skilled; a 0.92 efficiency is a plant figure. Both assumptions.
Cutting-tool wear	£0.14	£0.01	+£0.13	Same rate (tool-wear.aluminium £0.04/min). A charges it on 3.55 "cutting" min — which include the fixed approach allowances; B on 0.24 min of chip-making.	CF-20 (new)	B — Tools do not wear during positioning.
Machining setup, fixtures, CNC programming	£0.04	£0.03	+£0.01	A 3 hydraulic fixtures (one a machine the capacity implies) + £175 programming; B 2 fixtures (OP10 + OP20). Setup per part equal.	CF-05	B — B has 2 clampings (OP10/OP20) — the geometry needs them (probe 3); A costs 1.
Deburr / wash after machining (AvA)	£0.50	£0.24	+£0.26	A 0.5 + 0.004 × faces + 0.5 gauge ≈ 1.1 min ÷ 0.92; B 30 s + £0.03 wash.	CF-16	neither — Both unsourced.
Scrap adjustments (casting-scrap secondary, 1 % machining scrap)	£0.00	£0.06	−£0.06	A charges no machining scrap and no secondary spent on rejected castings; B 1 % machining scrap + 6 % casting scrap on fettle/HT/blast.	CF-09	B — Scrap at machining is real; B's rates are assumptions.
Packaging (AvA)	£0.09	£0.10	−£0.01	A £0.09 (rule by size); B £0.10 assumption.	—	neither — Both unsourced.
Overhead 12 %	£0.98	£0.82	+£0.16	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Margin 8 %	£0.74	£0.62	+£0.12	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Total	**£9.98**	**£8.36**	**+£1.62**			

Part1 (triple clamp) — A £13.82 v B £11.87: +£1.95 (+16.4 %)

- A as run before (gravity LM25 T6, 1 impression = the tool default): £14.6049.
- A with B's 2 impressions (casting.impressions 2): £13.8192 — used below.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Metal bought (net of melt loss / incl. bar kerf)	£4.13	£4.39	−£0.26	A as-cast 1.348 kg (1.5 mm stock) v B 1.399 kg (ISO 8062 for a 270 mm part: RMA E 1.8 + CT8/4 = 2.35 mm on 5 faces + cored-bore stock).	CF-02	B — ISO 8062 RMA grows with the largest dimension; A keeps 1.5 mm for any gravity casting.
Scrap credits (swarf, flash, wad, crop)	£0.00	−£0.14	+£0.14	A none; B 0.249 kg swarf × £0.54.	CF-14	B — Swarf is sold.
Melt / billet-heating energy + foundry indirect (melt-shop fixed, sand, bentonite, sand plant) (AvA)	£0.27	£0.26	+£0.02	As the bracket: A electric melt + plant kWh on 2.32 kg poured; B gas melter + holding + fixed £20/t + treatment on 2.98 kg poured (yield 0.5 v 0.6).	CF-10	neither — Yield and melt-shop fixed cost are assumptions on both sides.
Casting / forge cell (machine + crew, incl. melt crew) (AvA)	£0.89	£1.27	−£0.37	A die cycle 3.71 min / 2, crew 0.25; B 270 s / 2, crew 0.5 + 1/6 melt crew.	CF-16	neither — Assumptions both sides.
Primary tooling (die / pattern, maintenance, die change) (AvA)	£0.22	£0.21	+£0.00	A die £35,853 × 6; B £33,104 × 6 + maintenance — close.	—	neither — Agree within £0.01.
Fettling (AvA)	£0.92	£0.43	+£0.49	A 2 min flat; B 45 s.	CF-16	B — A flat 2 min unsourced.
Heat treatment (AvA)	£0.55	£0.88	−£0.33	A £0.41/kg station ESTIMATE v B £0.63/kg furnace hours.	CF-13	neither — No toll price.
Shot blast / descale (AvA)	£0.11	£0.06	+£0.05	A heuristic throughput; B 10 min / 150 kg.	CF-13	neither — Both unsourced.
Inspection and NDT	£0.00	£0.33	−£0.33	A none; B visual + X-ray audit + gauging + CMM.	CF-12	B (partly) — As the bracket.
Machining — non-cutting time: A fixed per-feature allowances + handling v B per-operation non-cut (AvA)	£2.64	£1.05	+£1.59	A: 4.55 min of fixed allowances (0.2 min a face, 0.15 a hole, 0.25 a bore pass, 0.1 blind, 0.1 tap; "engineering-typical, not sourced") + 0.12 min handling / pallet change = 4.67 min; B: 2.30 min non-cut (tool changes, approaches, probing, indexing, pallet change). A at B's OEE / operator basis. A 6.02 min v B 3.35 min (1.05 cut + 2.30 non-cut).	CF-19 (new)	B (more plausible) — Both are assumptions; B builds non-cut per operation from tool changes, approaches, probing and indexing; A adds 9–15 s per feature. Neither is a timed cycle.
Machining — cutting time (feature set, passes, cutting data)	£0.62	£0.48	+£0.14	A 1.35 min cutting (incl. 2.2 s tool changes and any lathe op) v B 1.05 min, both at B's basis, split by minutes. A 14 faces (22,761 mm²) v B 5 faces (11,663 mm²); B also cuts the two 3 mm slits with a saw.	CF-01, CF-06	neither — As the bracket.
Machining — OEE 0.8 on machine time (A) v none (B) (AvA)	£0.09	£0.00	+£0.09	A divides machine time by OEE 0.8 (core.ts:247); B applies no availability / performance loss.	CF-16	A — A dedicated cell still loses availability; B is optimistic. A's 0.8 is unsourced.
Machining — operator grade and efficiency (AvA)	£0.07	£0.00	+£0.07	A: skilled machinist £27.95 ÷ 0.92 efficiency, 0.5 crew; B: semi-skilled £24.65 at 100 %, 0.5 crew.	CF-16	neither — A production-cell loader is semi-skilled; a 0.92 efficiency is a plant figure. Both assumptions.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Cutting-tool wear	£0.24	£0.04	+£0.19	Same £0.04/min; A on 5.9 min incl. allowances, B on 1.05 min.	CF-20 (new)	B — Wear on positioning time.
Machining setup, fixtures, CNC programming	£0.07	£0.04	+£0.02	A 5 fixtures (4.18 HMCs) in 1 clamping; B 3 fixtures (2 OP10 machines + 1 OP20).	CF-05	B — Part1 needs ≥ 2 clampings (probe 3).
Deburr / wash after machining (AvA)	£0.52	£0.30	+£0.22	A ~1.1 min ÷ 0.92; B 40 s + wash.	CF-16	neither — Both unsourced.
Scrap adjustments (casting-scrap secondary, 1 % machining scrap)	£0.00	£0.09	−£0.09	A none; B 1 % machining scrap + casting scrap on secondary.	CF-09	B
Packaging (AvA)	£0.10	£0.12	−£0.02	A £0.10; B £0.12.	—	neither — Both unsourced.
Overhead 12 %	£1.36	£1.16	+£0.20	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Margin 8 %	£1.02	£0.88	+£0.14	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Rounding (B total printed to 3 dp; 4-dp rounding of the rows)	£0.00	£0.00	+£0.00	arithmetic	—	n/a
Total	**£13.82**	**£11.87**	**+£1.95**			

PRCR002 (stub axle) — A £25.79 v B £16.20: +£9.59 (+59.2 %)

- Same route and grade (sand, EN-GJS-500-7, DISA-class vertical line). A re-run with casting.processChoice sand + material.grade mat-gjs500 gives the identical £25.7885.
- Impressions differ and cannot be answered on the sand route: A 4 a mould (60 % plate rule), B 2 (assumption). Cores: A band £3.00 (32 undercut faces), B 2 shell cores built up.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Metal bought (net of melt loss / incl. bar kerf)	£3.41	£3.56	−£0.16	A as-cast 9.084 kg (3.0 mm on 8 face rows + cored bores + 3 mm turning stock on the 226 cm² spindle) + 2 % loss on returns → 9.261 kg × £0.368; B raw 9.278 kg (3.6 mm = RMA F 2.5 + CT10/4 on faces + spindle) + 2 % remelt loss + inoculant £0.10.	CF-02	B — Size-dependent ISO 8062 allowance; B also prices the inoculant A leaves out.
Scrap credits (swarf, flash, wad, crop)	£0.00	−£0.29	+£0.29	A none; B 1.91 kg swarf × £0.15.	CF-14	B — Iron borings are sold.
Melt / billet-heating energy + foundry indirect (melt-shop fixed, sand, bentonite, sand plant) (AvA)	£2.71	£3.03	−£0.32	A: melt + plant electricity (455 kWh/t good, ESTIMATE) + ladle gas + dust on 14.98 kg poured (yield 0.625) + new sand £11/t = £2.71; B: 620 kWh/t (sourced range) on 16.28 kg poured (yield 0.6) + £30/t fixed + sand 5:1 × 5 % + bentonite + £15/t sand plant = £3.03.	CF-10	neither — B's fixed £/t, sand ratio and bentonite are assumptions; A's plant electricity is a grade-D estimate.
Casting / forge cell (machine + crew, incl. melt crew) (AvA)	£0.82	£0.52	+£0.29	A DISA line 1/280 moulds/h ÷ 4 impressions (60 % plate rule), crew 3, + melt shop 1.5 operator-h/t (£0.60); B 300 moulds/h × 2 impressions, crew 8 covering melt, pour, cores, shake-out.	CF-16	neither — A's impressions follow the DISA plate rule (sourced); B's 2 impressions and crew 8 are assumptions; A's melt crew is heavier.
Cores	£3.09	£0.32	+£2.77	A £3.00 band for "32 undercut faces — complex multi-core" × 1.03; B 2 shell cores Ø30.8 × 100 mm built up (0.224 kg × £0.45/kg + 2 × 10 s shooting).	CF-11	B — B is a build-up from the cored features; A is an unsourced band keyed to a face count. B's inputs are assumptions too.
Primary tooling (die / pattern, maintenance, die change) (AvA)	£0.05	£0.06	−£0.01	A pattern £16,483 × 3 sets; B £19,058 × 3 — close.	—	neither — Agree.
Fettling (AvA)	£0.44	£0.36	+£0.08	Both 40 s on the robot cell; A × reject uplift and OEE 0.8.	—	neither — Same assumption (Malleable Castings time study).
Shot blast / descale (AvA)	£0.75	£0.29	+£0.46	A 9.08 kg ÷ 600 kg/h × (machine + full operator); B ÷ 1,000 kg/h, crew 0.25.	CF-13	neither — Both unsourced.
Inspection and NDT (AvA)	£0.77	£0.55	+£0.22	A MPI 75 s + RT 1 in 50 × £5 (safety-critical); B visual 10 s + MPI 30 s + ultrasonic 8 s + gauging 15 s + CMM audit.	CF-12	neither — MPI 75 s v 30 s — both assumptions; B adds in-process gauging A lacks.
Machining — non-cutting time: A fixed per-feature allowances + handling v B per-operation non-cut (AvA)	£3.18	£1.64	+£1.55	A: 4.51 min of fixed allowances (0.2 min a face, 0.15 a hole, 0.25 a bore pass, 0.1 blind, 0.1 tap; "engineering-typical, not sourced") + 1.32 min handling / pallet change = 5.83 min; B: 3.70 min non-cut (tool changes, approaches, probing, indexing, pallet change). A at B's OEE / operator basis. A 12.46 min (HMC faces/holes/bores + lathe spindle 1.72 + lathe load 1.2) v B 8.52 min (4.82 cut + 3.70 non-cut; rough + finish king-pin bores, OP30).	CF-19 (new)	B (more plausible) — Both are assumptions; B builds non-cut per operation from tool changes, approaches, probing and indexing; A adds 9–15 s per feature. Neither is a timed cycle.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Machining — cutting time (feature set, passes, cutting data)	£3.12	£2.13	+£0.99	A 6.63 min cutting (incl. 2.2 s tool changes and any lathe op) v B 4.82 min, both at B's basis, split by minutes. A includes the lathe spindle op (1.72 min) and one pass per face and bore; B turns the spindle between centres (1.90 min cut) and rough + finish bores the king-pins (OP10 + OP30).	CF-01, CF-06	B — B's rough + finish bore structure follows stock and tolerance (CF-06).
Machining — OEE 0.8 on machine time (A) v none (B) (AvA)	£0.46	£0.00	+£0.46	A divides machine time by OEE 0.8 (core.ts:247); B applies no availability / performance loss.	CF-16	A — A dedicated cell still loses availability; B is optimistic. A's 0.8 is unsourced.
Machining — operator grade and efficiency (AvA)	£0.44	£0.00	+£0.44	A: skilled machinist £27.95 ÷ 0.92 efficiency, 0.5 crew; B: semi-skilled £24.65 at 100 %, 0.5 crew.	CF-16	neither — A production-cell loader is semi-skilled; a 0.92 efficiency is a plant figure. Both assumptions.
Cutting-tool wear	£1.11	£0.48	+£0.63	Same rate (£0.10/min cast iron). A 11.1 "cutting" min incl. allowances; B 4.8 min of chip-making.	CF-20 (new)	B — Wear charged on positioning time.
Machining setup, fixtures, CNC programming	£0.11	£0.09	+£0.02	A 2 HMC + 1 lathe fixtures; B 3 HMC (OP10) + 3 lathe (OP20) + 2 HMC (OP30) fixtures.	CF-05, CF-08	B — B sequences OP10 / OP20 (between centres) / OP30 (king-pins from the turned spindle); A has 1 HMC clamping.
Deburr / wash after machining (AvA)	£0.54	£0.27	+£0.28	A ~1.2 min ÷ 0.92; B 30 s + wash / RP oil.	CF-16	neither — Both unsourced.
Scrap adjustments (casting-scrap secondary, 1 % machining scrap)	£0.00	£0.16	−£0.16	A none; B 1 % machining scrap £0.12 + casting scrap on secondary.	CF-09	B
Packaging (AvA)	£0.36	£0.25	+£0.11	A £0.36 (size rule); B £0.25.	—	neither — Both unsourced.
Overhead 12 %	£2.52	£1.58	+£0.94	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Margin 8 %	£1.91	£1.20	+£0.71	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Total	**£25.79**	**£16.20**	**+£9.59**			

Steering knuckle RH — A £12.53 v B £9.16: +£3.37 (+36.8 %)

- A as recorded = closed-die forging in 38MnVS6: £18.7363 (route note: B estimates forging at +£5–8 over its £9.16 casting, i.e. £14–17, without costing it).
- A re-run as cast + machine, EN-GJS-400-15 (material.grade mat-gjs400), sand, safety-critical yes: £12.5294 — used below. A still adds its near-net "spindle" lathe op (CF-07).

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Metal bought (net of melt loss / incl. bar kerf)	£1.08	£1.06	+£0.02	A as-cast 3.238 kg (3.0 mm sand stock on its 8 face rows and the cored Ø68 bore, plus turning stock on the 71 cm² "spindle") v B 3.097 kg (3.0 mm, RMA F 2.0 + CT10/4, 9 faces); same £0.328/kg + B inoculant.	CF-07	B — A adds turning stock for a lathe op the knuckle does not have (CF-07).
Scrap credits (swarf, flash, wad, crop)	£0.00	−£0.09	+£0.09	A none; B 0.57 kg swarf × £0.15.	CF-14	B
Melt / billet-heating energy + foundry indirect (melt-shop fixed, sand, bentonite, sand plant) (AvA)	£0.97	£1.01	−£0.05	A £0.97 on 5.34 kg poured; B £1.01 on 5.43 kg poured — close in total, split differently.	CF-10	neither
Casting / forge cell (machine + crew, incl. melt crew) (AvA)	£0.39	£0.52	−£0.14	A 5 impressions a mould (plate rule), crew 3 + melt shop 1.5 op-h/t; B 2 impressions, crew 8.	CF-16	neither — Assumptions both sides.
Cores	£0.77	£0.00	+£0.77	A £0.75 band for "1 undercut face — a small core or loose piece"; B no core (1 undercut face; bore formed by the mould halves).	CF-11	B — One undercut face does not justify a core; A's band fires on any count ≥ 1.
Primary tooling (die / pattern, maintenance, die change) (AvA)	£0.03	£0.04	−£0.01	A £11,639 × 3; B £14,627 × 3.	—	neither — Close.
Fettling (AvA)	£0.44	£0.36	+£0.08	Both 40 s robot cell.	—	neither — Same assumption.
Shot blast / descale (AvA)	£0.27	£0.10	+£0.17	A 600 kg/h + full operator; B 1,000 kg/h, crew 0.25.	CF-13	neither — Both unsourced.
Inspection and NDT (AvA)	£0.77	£0.55	+£0.22	A MPI 75 s + RT 1/50; B visual + MPI 30 s + US + gauging + CMM.	CF-12	neither — Both assumptions.
Machining — non-cutting time: A fixed per-feature allowances + handling v B per-operation non-cut (AvA)	£2.01	£1.33	+£0.68	A: 2.95 min of fixed allowances (0.2 min a face, 0.15 a hole, 0.25 a bore pass, 0.1 blind, 0.1 tap; "engineering-typical, not sourced") + 0.72 min handling / pallet change = 3.67 min; B: 2.92 min non-cut (tool changes, approaches, probing, indexing, pallet change). A at B's OEE / operator basis. A 7.37 min incl. a lathe "spindle" op 0.82 min + load 0.6 min (CF-07) v B 6.70 min (3.78 cut incl. rough + finish faces and an H7 bore, 2.92 non-cut).	CF-19 (new)	B (more plausible) — Both are assumptions; B builds non-cut per operation from tool changes, approaches, probing and indexing; A adds 9–15 s per feature. Neither is a timed cycle.
Machining — cutting time (feature set, passes, cutting data)	£1.73	£1.73	+£0.01	A 3.70 min cutting (incl. 2.2 s tool changes and any lathe op) v B 3.78 min, both at B's basis, split by minutes. Equal in total, but A's includes a lathe "spindle" op (0.82 min) the knuckle does not have (CF-07); on the HMC A cuts ≈ 2.9 min v B 3.8 min because B roughs + finishes faces and bores the H7 seat (CF-06).	CF-06, CF-07	B — A adds a lathe op the part does not need and omits B's rough / finish passes.
Machining — OEE 0.8 on machine time (A) v none (B) (AvA)	£0.26	£0.00	+£0.26	A divides machine time by OEE 0.8 (core.ts:247); B applies no availability / performance loss.	CF-16	A — A dedicated cell still loses availability; B is optimistic. A's 0.8 is unsourced.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Machining — operator grade and efficiency (AvA)	£0.24	£0.00	+£0.24	A: skilled machinist £27.95 ÷ 0.92 efficiency, 0.5 crew; B: semi-skilled £24.65 at 100 %, 0.5 crew.	CF-16	neither — A production-cell loader is semi-skilled; a 0.92 efficiency is a plant figure. Both assumptions.
Cutting-tool wear	£0.66	£0.38	+£0.29	Same £0.10/min; A 6.6 min incl. allowances and the lathe op; B 3.8 min.	CF-20 (new), CF-07	B
Machining setup, fixtures, CNC programming	£0.08	£0.09	−£0.01	A 4 HMC + 1 lathe fixture; B 4 + 2 HMC fixtures (OP10, OP20).	CF-05	B — B has 2 clampings; the geometry needs ≥ 2 (probe 3).
Deburr / wash after machining (AvA)	£0.51	£0.26	+£0.25	A ~1.1 min ÷ 0.92; B 30 s + wash.	CF-16	neither
Scrap adjustments (casting-scrap secondary, 1 % machining scrap)	£0.00	£0.10	−£0.10	A none; B 1 % machining scrap + casting scrap on secondary.	CF-09	B
Packaging (AvA)	£0.16	£0.15	+£0.01	A £0.16; B £0.15.	—	neither
Overhead 12 %	£1.23	£0.89	+£0.33	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Margin 8 %	£0.93	£0.68	+£0.25	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Total	**£12.53**	**£9.16**	**+£3.37**			

Hub flange (MODELLED) — A £13.37 v B £10.44: +£2.93 (+28.1 %)

- MODELLED part. Both closed-die 38MnVS6, controlled cooled, no heat treatment. A: 1,600 t press, trim 100 t, 2 hits; B: 2,500 t press, trim + pierce 400 t, takt 7.5 s. A answers safetyCritical "no" → no MPI; B charges 100 % MPI (an assumption for a wheel-end part) — the tool offers the answer, not changed so A stays as recorded.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Metal bought (net of melt loss / incl. bar kerf)	£5.64	£4.79	+£0.85	A billet = as-forged 3.185 kg ÷ 0.675 (unsourced band; 25.7 % of the billet "scale, crop, wad" not itemised) = 4.716 kg + kerf → 4.903 kg bar; B billet 3.947 kg = raw 3.425 + flash 0.348 (20 × 3 mm land, assumption) + wad 0.145 + scale 0.75 % + excess 3 % → 4.10 kg bar. Same £1.15/kg.	CF-04	B — B itemises every loss term; A's yield band is unsourced and constant. B's flash land, scale % and excess are still assumptions. Note A's as-forged weight is LOWER (3.185 v 3.425 kg) — B adds EN 10243-1 minus tolerance and 3°/5° drafts; it is A's billet ratio (1.48 v 1.15) that drives the gap.
Scrap credits (swarf, flash, wad, crop)	−£0.33	−£0.39	+£0.06	A credits flash + un-itemised loss + kerf (1.65 kg × £0.20) but no swarf; B credits flash, wad, kerf, crop (0.71 kg) and 1.24 kg swarf, not scale.	CF-14, CF-04	B — Scale is not recoverable; swarf is.
Melt / billet-heating energy + foundry indirect (melt-shop fixed, sand, bentonite, sand plant)	£0.35	£0.29	+£0.06	Same 0.4 kWh/kg (SMS Elotherm); A heats a 4.72 kg billet, B 3.95 kg.	CF-04	B — Consequence of the billet mass.
Casting / forge cell (machine + crew, incl. melt crew) (AvA)	£0.88	£0.66	+£0.21	A 1,600 t press, takt 3 s + 2 hits × 4 s = 11 s, crew 2 + trim 1, OEE 0.8; B 2,500 t press, takt 7.5 s, crew 3, no OEE.	CF-16	neither — Takt and crew unsourced on both sides; A picks a smaller press, B a faster takt.
Primary tooling (die / pattern, maintenance, die change) (AvA)	£0.53	£0.60	−£0.07	A die set £25,447 ÷ 47,955 forgings (fractional); B insert pairs by life + holder + die change £0.06.	CF-16	neither — Both build-ups are judgement.
Shot blast / descale (AvA)	£0.39	£0.09	+£0.30	A £0.12/kg × forging (unsourced); B 1,200 kg/h blast.	CF-13	B — A's £/kg is ~4× B's build-up; neither sourced.
Inspection and NDT	£0.00	£0.36	−£0.36	A none (safety-critical answered "no" in ANSWERS); B 100 % MPI 25 s (its assumption for a wheel-end part) £0.22 + gauging + CMM £0.14.	CF-12 (gauging / CMM); MPI = a route choice	B (gauging / CMM) — The MPI is an answer difference, not method: the modelled part is stated not safety-critical.
Machining — non-cutting time: A fixed per-feature allowances + handling v B per-operation non-cut (AvA)	£1.32	£0.69	+£0.62	A: 1.15 min of fixed allowances (0.2 min a face, 0.15 a hole, 0.25 a bore pass, 0.1 blind, 0.1 tap; "engineering-typical, not sourced") + 1.20 min handling / pallet change = 2.35 min; B: 1.72 min non-cut (tool changes, approaches, probing, indexing, pallet change). A at B's OEE / operator basis. A 4.11 min (lathe op10/op20 1.45 + 1 load 0.6; holes + Ø60 bore on a 5-axis 1.46 + load 0.6) v B 4.08 min (two lathe chuckings + VMC drilling).	CF-19 (new)	B (more plausible) — Both are assumptions; B builds non-cut per operation from tool changes, approaches, probing and indexing; A adds 9–15 s per feature. Neither is a timed cycle.

Driver	A £	B £	Gap £	Cause	Gap id	Who looks right
Machining — cutting time (feature set, passes, cutting data)	£0.91	£0.95	−£0.04	A 1.76 min cutting (incl. 2.2 s tool changes and any lathe op) v B 2.36 min, both at B's basis, split by minutes. A turns outside + faces in one lathe op (1.45 min) and drills / bores on a 5-axis; B turns op 10 and op 20 (2.07 min cut) and drills on a VMC.	CF-06, CF-07	B — B's op 10 / op 20 structure holds the bore-to-OD datum (CF-08).
Machining — OEE 0.8 on machine time (A) v none (B) (AvA)	£0.17	£0.00	+£0.17	A divides machine time by OEE 0.8 (core.ts:247); B applies no availability / performance loss.	CF-16	A — A dedicated cell still loses availability; B is optimistic. A's 0.8 is unsourced.
Machining — operator grade and efficiency (AvA)	£0.20	£0.00	+£0.20	A: skilled machinist £27.95 ÷ 0.92 efficiency, 0.5 crew; B: semi-skilled £24.65 at 100 %, 0.5 crew.	CF-16	neither — A production-cell loader is semi-skilled; a 0.92 efficiency is a plant figure. Both assumptions.
Cutting-tool wear	£0.35	£0.28	+£0.07	Same £0.12/min; A 2.9 min incl. allowances; B 2.36 min.	CF-20 (new)	B
Machining setup, fixtures, CNC programming	£0.03	£0.05	−£0.02	A lathe fixture + 1 5-axis fixture; B 2 + 2 soft-jaw sets + 1 VMC fixture + jaw replacement.	CF-08	B — B turns op 10 / op 20 in two chuckings; A charges one.
Deburr / wash after machining	£0.46	£0.05	+£0.41	A ~1.04 min ÷ 0.92; B wash only (chamfers in OP30).	CF-16	A — B charges no deburr of 6 drilled holes — optimistic.
Scrap adjustments (casting-scrap secondary, 1 % machining scrap)	£0.00	£0.08	−£0.08	A none; B 1 % machining scrap.	CF-09	B
Packaging (AvA)	£0.18	£0.12	+£0.06	A £0.18; B £0.12.	—	neither
Overhead 12 %	£1.31	£1.02	+£0.28	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Margin 8 %	£0.99	£0.77	+£0.22	Same rate on both sides (12 % of the factory base; 8 % of the subtotal incl. packaging) — the gap is the rate × the other gaps.	follows	n/a
Total	**£13.37**	**£10.44**	**+£2.93**			

Where B looks wrong or optimistic

- No OEE anywhere in B (casting cells, press and machining at 100 %).
- Bracket machining 1.74 min with 0.24 min cutting; B's non-cut (tool changes 3 s, approaches 2 s, probing 12 s, B-index 2 s) is its own assumption, swung ×0.7 / ×1.4 in its range. B machines 4 faces on the bracket; A 8. Neither choice is backed by a drawing.
- Foundry inputs are assumptions: 60 % ductile-iron yield, melt-shop fixed £30/t, sand plant £15/t, bentonite £0.01/kg, sand 5 : 1, crew 8 on the moulding line.
- X-ray audits on non-safety aluminium parts and 100 % MPI on a hub flange stated not safety-critical (£0.22 — a route choice, not method).
- Hub flange: no deburr of 6 drilled holes. CT8 assumed for permanent mould (ISO 8062 gives none); EN 10243-1 tolerance estimated, the table was not read.

Historic CH-47 spiral bevel gear (TRW 1972 record v the tool)

> Hours and counts only — no currency. The record is 1972 standard hours on 1972 US machines; the tool is 2026 CNC cycle times with OEE 0.8, efficiency 0.92 and crew 0.5. The STEP models are MODELLED FROM THE PUBLISHED DRAWING (straight-skew teeth, face width 1.00 in assumed). A ratio mixes era, method and scope; it is not an accuracy figure for either side.

Comparison	Recorded	Tool	Recorded ÷ tool
Forged route, labour h/pc, Table XXIX basis	1.514 h	0.110 h	13.8×
Forged route, full Table XXV (DERIVED)	2.444 h	0.110 h	22.3×
Conventional route, labour h/pc, Table XXIX	2.272 h	0.037 h	61×
Forged billet	7.26 kg (16 lb, Ø88.9 mm bar)	7.91 kg (Ø82.3 mm bar)	tool +9 %
Set-ups	5 forging per 500-lot (15.0 h); 11 machining per 100-lot (19.0 h)	0 forging; 1 amortised machining line of 1.5 h per batch	different in kind

- No bevel-gear capability: gear metrology reads nothing on a bevel gear; the gear module models a cylindrical helical gear (tip Ø 254.7 mm v 226.2 mm heel OD); the forging route picks a ring mill; with neutral file names the gear is routed only to casting / cast + machine / machining (CF-21).
- Every pre-heat-treat grinding operation (4 gear, 1 pinion), polishing and marking are absent; 7–9 lathe operations collapse into 2 lines (CF-22, CF-06).
- The +9 % billet agreement is partly coincidental: reached through a ring-rolling yield of 0.725 on 3 mm stock, not closed-die practice (the drawing calls 0.15–0.25 mm on the teeth).

14. Single-gap sensitivities (probes P4 / P6, UK, 50k/yr, baseline answers)

Each row changes one input of the tool and re-costs through the same chain; the repository is not modified (tables mutated in memory). These are sensitivities of the tool, not corrections.

Gap	Change	Part	New total	Δ £	Δ %
CF-02	Casting stock 3.0 → 1.5 mm a side	Casting bracket (steel sand, 50k)	£10.40	−£0.31	−2.9 %
CF-02	Casting stock 3.0 → 6.0 mm a side	Casting bracket (steel sand, 50k)	£11.35	+£0.63	+5.8 %
CF-02	Casting stock 3.0 → 1.5 mm a side	PRCR002 (50k)	£26.56	−£0.88	−3.2 %
CF-02	Casting stock 3.0 → 6.0 mm a side	PRCR002 (50k)	£29.22	+£1.77	+6.5 %
CF-02	Forging stock 2.5 → 1.5 mm a side	Knuckle forging (50k)	£19.08	−£0.38	−2.0 %
CF-02	Forging stock 2.5 → 4.0 mm a side	Knuckle forging (50k)	£20.06	+£0.59	+3.0 %
CF-01	Only the 4 eye faces machined (design intent) instead of 22 planar faces	Yoke (MODELLED, 50k)	£10.18	−£6.66	−39.5 %

Gap	Change	Part	New total	Δ £	Δ %
CF-07	No lathe "spindle" operation on the eye ODs	Yoke (MODELLED, 50k)	£15.65	−£1.19	−7.0 %
CF-05	2 clampings on the HMC cell instead of 1	Casting bracket (50k)	£11.10	+£0.39	+3.6 %
CF-05	2 clampings on the HMC cell instead of 1	PRCR002 (50k)	£28.12	+£0.68	+2.5 %
CF-05	2 clampings on the HMC cell instead of 1	Knuckle forging (50k)	£20.44	+£0.97	+5.0 %
CF-05	2 clampings on the HMC cell instead of 1	Part1 (machining) (50k)	£40.08	+£1.27	+3.3 %
CF-10	Casting yield 0.525 → 0.40	Casting bracket (50k)	£11.16	+£0.44	+4.1 %
CF-10	Casting yield 0.625 → 0.40	PRCR002 (50k)	£29.40	+£1.96	+7.1 %
CF-04	Closed-die yield band 0.675 → 0.55	Knuckle forging (50k)	£21.16	+£1.70	+8.7 %
CF-04	Closed-die yield band 0.675 → 0.80	Knuckle forging (50k)	£18.31	−£1.16	−6.0 %
CF-04	Closed-die yield band 0.675 → 0.80	Hub flange (MODELLED, 50k)	£12.89	−£0.96	−6.9 %
CF-07	No lathe "spindle" operation	Knuckle forging (50k)	£18.11	−£1.35	−6.9 %
CF-06	A second boring pass (approximated by halving the boring feed)	PRCR002 (50k)	£27.70	+£0.26	+0.9 %
CF-06	A second boring pass (approximated by halving the boring feed)	Knuckle forging (50k)	£19.86	+£0.39	+2.0 %
CF-09	2 % machining scrap (estimate from the factory base, not a re-cost)	Casting bracket (50k)	estimate	+£0.21	+2.0 %
CF-09	2 % machining scrap (estimate from the factory base, not a re-cost)	PRCR002 (50k)	estimate	+£0.54	+2.0 %

15. Prioritised corrective actions

Priority by the brief's Phase 10 scale. The implementation packages are set out in deliverable B (Recommended Methodology); none is implemented — approval is required first.

Id	Priority	Gap	Quantified impact (from probes)	Package
CF-01	Critical	Machined surfaces are a geometric heuristic (8 largest planar areas + every hole), not the drawing	Yoke £16.83 → £10.18 (−39.5 %) when only the eye faces are machined (machining-centre min 9.18 → 3.17; as-forged 2.316 → 1.818 kg). On the genuine parts the error cannot be measured without their drawings.	PKG-1
CF-02	Critical	Machining allowance is one uniform per-process constant; no RMA/DCTG, no size, no per-feature, no roughing/finishing split	Per ±1.5 mm of stock: bracket ±£0.31 (±2.9 %), PRCR002 ±£0.88 (±3.2 %); knuckle forging −£0.38/+£0.59 (−2.0/+3.0 %) for 1.5/4 mm. Machining time does not move at all (bracket HMC 4.73 min at 1.5–6 mm), so the cutting cost of extra stock is never charged.	PKG-1
CF-03	Critical	No raw-to-finished model: one finished CAD + one drawing; no drawing pair, no raw CAD, drawing dimensions / GD&T / Ra extracted but never costed	Not quantifiable without genuine raw + finished pairs; it is the root cause of CF-01, CF-02 and CF-06.	PKG-1
CF-04	Critical	Forging material balance: billet = as-forged ÷ 0.675 (unsourced) with flash fixed at 10 %; 25.7 % of every billet is un-itemised "scale, crop, wad"	Closed-die band 0.55 / 0.80 / 0.90: knuckle +£1.70 (+8.7 %) / −£1.16 (−6.0 %) / −£1.86 (−9.6 %); hub +£1.41 (+10.2 %) / −£0.96 (−6.9 %) / −£1.54 (−11.1 %).	PKG-4
CF-05	High	Setup / clamping count: kernel caps directions at +X/+Y/+Z from ALL faces; HMC cell assumes ≤ 4 directions in one clamping; 5-axis one setup	P4-D (2 clampings): bracket +£0.39 (+3.6 %), PRCR002 +£0.68 (+2.5 %), knuckle +£0.97 (+5.0 %), Part1 (machining) +£1.27 (+3.3 %); fixtures double.	PKG-3
CF-06	High	Machining time independent of stock and tolerance: one finish pass per face and bore; no semi-finish / fine bore / ream / grind / hone	Second bore pass ≈ +£0.26 (PRCR002), +£0.39 (knuckle); grinding/honing of bearing seats not quantified (needs drawing tolerances).	PKG-2

Id	Priority	Gap	Quantified impact (from probes)	Package
CF-07	High	Near-net "spindle" lathe operation fires on parts with no spindle (knuckle 9 %, yoke 9.1 % ≥ 8 % threshold)	P6 knuckle not turned: £19.47 → £18.11 (−£1.35, −7.0 %); P4-C2 yoke −£1.19 (−7.0 %).	PKG-1
CF-08	High	No operation sequence or datum scheme; per-side ops apportioned by ALL-face counts; datum-related features split across machines	Hub flange: missing second lathe load ≈ £0.46 before overhead/margin (≈ +£0.55, +4 %); the per-side split misattributes cost but does not change the total.	PKG-3
CF-15	High	Machine selection does not justify axes from geometry; 5-axis assumed one setup; part weight / tolerance capability not checked	Included in CF-05 quantification (+2.5 – 5 %); ranking can flip when the true setups are used.	PKG-3
CF-19	High	Non-cutting time is a fixed approach allowance per feature (0.06–0.5 min, unsourced) — 28–76 % of machining minutes	A's fixed allowances are £1.42 / £2.58 / £2.56 / £1.68 / £0.70 a part (bracket / Part1 / PRCR002 / knuckle / hub, pre-overhead; £8.94 in total). Net of B's own per-operation non-cut, A's non-cutting time is £0.79 / £1.59 / £1.55 / £0.68 / £0.62 a part higher (£5.23 in total, ×1.21 with overhead and margin): A non-cut 2.74 / 4.67 / 5.83 / 3.67 / 2.35 min v B 1.50 / 2.30 / 3.70 / 2.92 / 1.72 min.	PKG-2
CF-21	High	No bevel / spiral-bevel gear capability: gear metrology reads nothing, the helical model gives the wrong OD, the forging route picks a ring mill	Recorded / tool labour hours 14x (forged route, Table XXIX basis) and 61x (conventional route); era, method and scope sit inside those ratios; not convertible to GBP.	PKG-7
CF-09	Medium	Machining scrap not costed on cast + machine, casting or forging secondary machining	Estimated +2.0 %: bracket +£0.21, PRCR002 +£0.54 (2 % of factory base, overhead + margin on it — P6, an estimate, not a re-cost).	PKG-4
CF-10	Medium	Casting yield (gating / risers) is a band midpoint, not a feeding design	P4-E yield 0.40 / 0.75: bracket +£0.45 / −£0.43 (±4 %); PRCR002 +£1.96 (+7.1 %) / −£0.53 (−1.9 %).	PKG-5
CF-11	Medium	Core cost is a band by undercut-face count, unrelated to the cored-bore rule	Not quantifiable without a core-shop quote (confirms the CLAUDE.md statement that core cost is not built up).	PKG-5
CF-12	Medium	Inspection is a 0.5-min gauge check plus NDT only when safety-critical; no CMM / SPC / probing / tolerance-driven inspection	Not quantified (needs inspection plan / tolerances).	PKG-5
CF-13	Medium	Secondary-operation gaps: steel riser cut-off / weld repair not costed; unsourced £/kg services carry a second overhead + margin	Bought-in services × 20.96 % (12 % then 8 %): bracket heat treat + blast ≈ £1.32 → ≈ £0.28 of overhead/margin on them; riser cut-off not quantified.	PKG-5
CF-14	Medium	Scrap credits inconsistent: swarf never credited; scale credited at steel scrap price; rejected forgings not credited	−1 % to −1.5 % (swarf); scale over-credit small (unknown scale %).	PKG-4
CF-16	Medium	Setup / handling / allowance / programming / fixture constants unsourced; no probing, tool-life change or first-off time of its own	Not quantified; at 50k/yr setup is 1.1 s a part — cycle allowances dominate (e.g., 0.15 min a hole).	PKG-2
CF-20	Medium	Cutting-tool wear is charged on minutes that include the non-cutting allowances	A − B tool wear: +£0.13 bracket, +£0.19 Part1, +£0.63 PRCR002, +£0.29 knuckle, +£0.07 hub (part of it is content difference).	PKG-2
CF-22	Medium	Forge-line die setting and per-operation set-ups are not costed; pre-heat-treat grinding, polishing and marking are absent	Per-piece set-up 13-63x lower in the tool than in the 1972 record (era and lot size included).	PKG-2, PKG-7
CF-17	Low	Cost buckets mix material with energy, services, inspection and tool wear	Presentation only; totals unaffected.	PKG-6
CF-18	Low	Reporting inconsistencies: "near-net (no extra material)"; per-side op names; forging turning stock 2 mm v IS 3469 2.5 mm	Presentation; turning-stock inconsistency small.	PKG-6

16. Final independent assessment (brief Phase 12)

#	Question	Answer	Evidence
1	Does the application correctly understand as-cast / as-forged and final machined drawings?	**No.**	One drawing per costing (server/routes/cad.ts:773-776); no stage concept; dimensions, GD&T and Ra are read but never reach the cost (engineering-data packs bind material, tolerance class, safety, pressure-tight, finish only). Of the 4 genuine pairs found, the Stirling "raw" drawings are cast-part detail drawings, not pure as-cast drawings, and no pair has CAD.
2	Can it reliably associate the two drawings with the same component?	**No.**	It cannot take two drawings. Part number and revision are extracted from the title block but never compared with the CAD.
3	Are machining allowances calculated correctly?	**No.**	One per-process constant on every machined face (casting 0.5–3.0 mm; forging IS 3469 by size). B's ISO 8062 values differ (1.55 / 2.35 / 3.6 / 3.0 mm a side). ±1.5 mm moves cost ≈ ±3 %; machining minutes do not move at all.
4	Does it identify all required machining features and operations?	**Partly.**	Holes, stepped bores and lands are recognised well. Machined faces are the 8 largest planar areas (22 faces on the modelled yoke v 4 intended). No reaming, fine boring, grinding, honing or polishing; the historic gear shows every pre-heat-treat grinding operation missing.
5	Is the operation sequence technically feasible?	**No sequence exists.**	Operations are one per kernel direction plus drilling / boring / handling / deburr, with no OP10/OP20 or datums. 4 of 6 parts need ≥ 2 clampings on a single rotary axis but are costed in 1; the hub flange bore is cut away from the OD it must run true to.
6	Is CNC machine and axis selection justified?	**Partly.**	Three routings (3-axis split, 5-axis, twin-pallet HMC) are priced and the cheapest chosen, but on setup counts the geometry cannot achieve; 5-axis is assumed one setup even with opposite machined faces.
7	Are setup counts, fixture requirements and cycle times realistic?	**No.**	Setup count ≤ 3 by construction (kernel merges ±N and keeps +X/+Y/+Z). Fixture sets by machine capacity are reasonable. Fixed per-feature allowances are 28–76 % of machining minutes (P10); A's non-cut (allowances + handling) is 2.35–5.83 min a part v B's 1.50–3.70 min; A's cutting minutes are above B's on 3 parts and equal or below on 2. Historic gear: recorded ÷ tool labour 14× (forged) and 61× (conventional).
8	Are casting and forging costs separated correctly from machining costs?	**Yes in the logic; not in the presentation.**	Separate casting / forging modules and per-operation machining lines; no double counting found. The "raw material" bucket mixes metal with energy, heat treatment, NDT, cores and tool wear (metal is 24–29 % of it on the iron parts).
9	Are material consumption, scrap and recovery calculated correctly?	**Partly.**	Casting balance closes (bought = as-cast + melt loss on returns). Forging billet = as-forged ÷ 0.675 (unsourced), leaving 25.7 % of every billet un-itemised; swarf never credited; scale credited; machining scrap dropped. Metal net of credits is within −£0.12 to +£0.91 of B (A − B).
10	What is the quantified impact of the identified gaps?	**A is above B on all five parts: Casting bracket +£1.62 (+19.4 %); Part1 + £1.95 (+16.4 %); PRCR002 +£9.59 (+59.2 %); Steering knuckle RH +£3.37 (+36.8 %); Hub flange + £2.93 (+28.1 %).**	Σ gap £19.46: secondary-operation constants +£6.66, non-cutting time + £5.23 (A's fixed allowances alone are £8.94 of its cost), OEE / operator + £2.06, cutting time +£1.45, tool wear +£1.31, material and energy + £0.85, B-only inspection and scrap −£1.08, overhead + margin +£3.36. Single-gap probes: machined-face set −39.5 % (modelled yoke); 2 clampings +2.5 to +5.0 %; stock ±3 % per 1.5 mm; forging yield −9.6 / +8.7 %. B is an independent engineering estimate, not a market price.
11	What must be corrected before presenting to the Cost Engineering Director?	**PKG-1, PKG-2, PKG-3 and PKG-4, then one measured comparison against real prices.**	Machined-surface set and allowance from the drawing (CF-01/02/03), the non-cutting time model (CF-19/20), setups and sequence (CF-05/08), the forging balance (CF-04); and at least 3 parts compared with supplier quotes or cycle sheets. Until then no accuracy claim can be made.

Readiness status: Suitable for controlled internal testing

Scope: casting + machining and forging + machining of prismatic / near-net parts, UK book, with the controls listed — not bevel or spiral-bevel gears, which are out of scope (CF-21).

Why not "Not validated". The calculation itself has been tested: the deterministic chain reproduces its recorded baseline to the penny, every material bucket re-adds (P7), overhead and margin are applied once, casting balances close, and on five parts every £ of the difference from an independent bottom-up estimate is attributed to a named driver and a gap id. The errors are known, directional (A is high by +16 % to +59 %, mainly through unsourced non-cutting allowances and secondary-operation constants) and partly offsetting. A cost engineer can see and override each one on the screen (answers, feature ticks, typed weights). That is what controlled internal testing needs.

Why not higher. Accuracy is not validated: no part has a real price; B is an estimate. Four Critical gaps remain (machined surfaces, allowance, raw-to-finished, forging balance), setups and cycle time are High gaps, and there is no bevel-gear capability. The tool is not fit for a Director demonstration of accuracy.

Controls for the internal testing: (1) an engineer answers route, grade, impressions and safety, and confirms the machined-feature ticks on the casting / forging form; (2) every output is labelled "uncalibrated" and used only internally; (3) each tested part is paired with a supplier quote, cycle sheet or purchase price, recorded in scripts/actuals/cad-actuals-input.csv; (4) bevel and spiral-bevel gears are excluded; (5) the readiness is reviewed after PKG-1 to PKG-4 and at least 3 real-price comparisons.

17. Evidence files (castforge/)

File	Content
audit/assessment.json, audit/gaps.json	19 areas with code / formula / evidence; 22 gaps with method, source, logic, acceptance.
audit/a-v-b.json, audit/a-v-b.md	A v B driver decomposition per part.
audit/probes/p1 … p11	Probe scripts and outputs: replay (P1), face rows (P2), setup directions (P3), sensitivities (P4, P6), rates (P5, P9), balance (P7), A v B (P8), machining split (P10), build (P11).
current/*.json	Every A run: conditions, operations, material, tooling, buckets, trace, provenance.
reference/*.json, *.md, summary.json	B per part: route, allowance, OP10/20/30, cost lines with formulas, sources S1–S12, assumptions, ranges.
dataset/register.json, not-used.json	4 genuine raw / machined pairs; sources rejected and why.
historic/recorded.json, comparison.json, comparison.md	TRW 1972 Tables XXV / XXVI transcribed; tool runs; ratios and caveats.